PHYS 4041/5041
Fall 2023
Homework 5
Due: In class, Friday, October 13, 2023

1. The heat equation describes the flow of thermal energy (heat) in materials. The temperature Ψ obeys:

$$\frac{\partial \Psi}{\partial t} = \frac{k}{c_p \rho} \nabla^2 \Psi,$$

where t is time, k is the thermal conductivity, c_p is the specific heat capacity, and ρ is the mass density.

(a) Assuming the problem is spherically symmetric, write the heat equation in spherical coordinates.

(b) Apply the technique of separation of variables, assuming

$$\Psi = \frac{u(r) f(t)}{r}.$$

Find the the resulting differential equations for u and f, and provide their general solution. Note: (1) By factoring r^{-1} out of u as we have, the spherically-symmetric Laplacian is simplified and the solution to the differential equation for u should be easy to recognize. (2) Here I would like you to choose the sign of the separation constant so that the solutions for $f(t)$ are exponentials that decay with increasing time.

(c) What is the form of the solution when the separation constant is zero?

(d) Let's apply this equation to the cooking of a turkey – maybe this will come in handy next month. Let's assume that our "turkey" is a sphere of radius $R = 0.2$ m and starts out thawed and at room temperature ($T_0 = 25^\circ$ C) and that the oven is set at $T_1 = 180^\circ$ C. In order to be quantitative, assume that turkey has the same thermal properties as water:

thermal conductivity:	$k = 0.90$ W/(m·K)
heat capacity:	$c_p \rho = 4.18 \times 10^6$ J/m^3.

The recipe described above implies an initial condition of $\Psi(r, 0) = T_0$ and a spatial boundary condition of $\Psi(R, t) = T_1$. Clearly, the solution for Ψ must be regular at $r = 0$. Physical considerations also rule out choices of the separation constant that give rise to increasing exponentials. What is the most general form of the solution subject to these restrictions?

(e) In order to more easily impose the boundary conditions, is useful to consider $\Phi \equiv \Psi - T_1$ so that $\Phi(R, t) = 0$. Note that $\Phi(r, t)$ also satisfies the heat equation. What are the restrictions now on the radial solution?

(f) By imposing the initial conditions, determine the complete solution $\Psi(r, t)$.

(g) Considering only the first non-vanishing time-dependent term, how long does it take to cook this turkey? Assume the turkey is "done" when the core temperature reaches 70° C.

2. Find the steady state in a square metal plate of side length a, if one side is held at a temperature $T = T_0$ and the remaining sides are held at $T = 0$. Treat this as a two-dimensional problem. In order to find the steady state solution, the time derivative in the heat equation can be set to zero. The result is the Laplace Equation, with no dependence on the thermal properties of the material.

3. Using the basis functions $u_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$, with $n = 1, 2, 3, \ldots$, expand

$$y(x) = \begin{cases} \frac{hx}{a} & x < a \\ \frac{h(L-x)}{L-a} & x > a \end{cases}$$

over the interval $[0, L]$. This will be useful for a later homework problem involving a plucked guitar string.

4. Be sure to read the hint at the bottom. For the function

$$f(x) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left(-\frac{x^2}{4\sigma^2}\right),$$

answer the following.

(a) Show that the normalization $f(x)$ is such that $\int_\infty^\infty |f(x)|^2 \, dx = 1$.

(b) The standard deviation of x is Δx, defined by

$$\Delta x^2 = \int_{-\infty}^{\infty} x^2 \, |f(x)|^2 \, dx.$$

Show that $\Delta x = \sigma$. Extra hint: you can differentiate the main hint with respect to α or k.

(c) Determine $g(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) \, e^{-ikx} \, dx$.

(d) Show that $\int_{-\infty}^{\infty} |g(k)|^2 \, dk = \frac{1}{2\pi}$, consistent with Parseval's theorem.

(e) The standard deviation of k is Δk, defined by

$$\Delta k^2 = 2\pi \int_{-\infty}^{\infty} k^2 \, |g(k)|^2 \, dk.$$

Find Δk. What is the product $\Delta k \, \Delta x$?

Hint: you may assume the following definite integral:

$$\int_{-\infty}^{\infty} \exp(-\alpha x^2) \, \exp(\pm ikx) \, dx = \sqrt{\frac{\pi}{\alpha}} \, \exp\left(-\frac{k^2}{4\alpha}\right).$$

5. Suppose that a wavefunction is given by

$$\Psi(x,t) = \begin{cases} 0 & t < 0 \\ \exp[i(kx - (E_R - i\Gamma/2)t/\hbar)] & t \geq 0 \end{cases},$$

where E_R and $\Gamma > 0$ are parameters.

(a) What is $|\Psi(x,t)|^2$ for $t \geq 0$? Note that this wavefunction could represent that of state (or particle) produced at time $t = 0$ which decays with a mean lifetime of $\tau = \hbar/\Gamma$.

(b) This wavefunction can be transformed the the energy domain via a Fourier Transform:

$$\Phi(x,E) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x,t)\, \exp[i(Et/\hbar)]\, dt.$$

What is $\Phi(x,E)$?

(c) What is $|\Phi(x,E)|^2$? Plot it. The parameter Γ is called the Full Width at Half Maximum (FWHM). Does this make sense? Note that $\Gamma\tau = \hbar$ is a particular case of the uncertainty principle that is an exact relation. It holds for states which decay according the exponential decay law (which is often the case, due to decay *rates* being constant). It implies that unstable states (or particles) necessarily have a finite energy width.

6. The Discrete Fourier Transform (DFT)is defined by

$$g_m = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f_k \exp(2\pi i\, km/N) \qquad m = 0, 1, 2, \ldots, N-1.$$

where function $f(x)$ is used only for discrete equally-spaced x values:

$$\begin{aligned} f_k &= f(x_k) \\ x_k &= \frac{2\pi k}{N} \qquad k = 0, 1, 2, \ldots, N-1. \end{aligned}$$

In class, we derived an orthogonality relation

$$\sum_{k=0}^{N-1} \exp(-2\pi i\, km/N) \exp(2\pi i\, kn/N) = N\delta_{m,n}$$

by recognizing that the sum consists of a geometric series; note that $n = 0, 1, 2, \ldots, N-1$ as well. We then found that the inverse transform is given by

$$f_k = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} g_m \exp(-2\pi i\, mk/N) \qquad m = 0, 1, 2, \ldots, N-1.$$

Assume that f_k and g_m are related by the above equations parts (a) and (b) below.

(a) Show that

$$\sum_{m=0}^{N-1} |g_m|^2 = \sum_{k=0}^{N-1} |f_k|^2,$$

which is a version of Parseval's Theorem for the DFT.

(b) If $\tilde{f}_k = f_{k-n}$, where n is any integer, show that the DFT of $\tilde{f}_k$ is given by

$$\tilde{g}_m = \exp(2\pi i\, nm/N) g_m.$$

Note that if the index of f_k, $\tilde{f}_k$, g_m, or $\tilde{g}_m$ is outside of the range $0, 1, 2, \ldots, N-1$ it is understood to be brought back into this range by adding or subtracting the necessary integer multiple of N (equivalent to assuming they are periodic with a period of N). This result shows that shifting f_k by a constant offset multiplies g_m by a linearly-varying phase.